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Suspicious Control Flow Detection

An HPE Juniper Networking Techpost

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HPE

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1 Suspicious Control Flow Detection

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Juniper enhanced the initial DDoS protection feature with Suspicious Control Flow Detection (SCFD). It provides deeper analysis within a given protocol or packet-type: a solution that addresses the need for more granular flow policing, supported from Junos OS 17.1R1

1.1 Introduction

Another potent feature integrated into the MX series platform and the MX Trio silicon is the DDoS protection. MX ddos-protection is enabled by default, and the feature is co-operated with the RE filter. A policer configured in the lo0 input filter will be downloaded to the Trio PFE (Packet Forwarding Engine) where it is executed before any DDoS (Distributed Denial of Service) policer functionality. The default mode of ddos-protection feature is supported on all MX platforms. It is also available on the PTX10000 series, and certain ACX7000 models.

The ddos-protection feature is configurable from the CLI and is designed to protect both the control and forwarding plane.

A substantial enhancement has been developed to get more of the MX Trio: the Suspicious Control Flow Detection (SCFD). With this granular flow policing, Juniper has elevated the initial DDoS protection feature to enable more comprehensive analysis within specific protocols or packet types.

Unlike basic DDoS, SCFD remains deactivated by default. Once enabled, flow protection monitors the specific contributors to the various aggregate flows. It dynamically tries to determine those that are “hostile” by virtue of their accounting for the majority of the aggregate packet rate. Suspicious flows are monitored within a specified time frame to record both the moments when they are flagged as suspicious and when these flows revert back to acceptable levels. A suspect flow in violation of its configured rate can be dropped, policed, or forwarded based on configuration. The latter case does not provide any inherent protection over basic DDoS, but the identification and logging of flows that were found to be in violation can be useful for later analysis and possible subsequent DDoS tuning.

Trio offers massive subscriber scale in terms of logical interfaces, IPv4 and IPv6 routes, and hierarchical queuing.

The design of MX Trio PFE that has ASIC performance and flexibility of FPGA allows it. In Trio chipset, multiple instructions can be executed to a packet before it leaves the forwarding engine; run-to-completion.

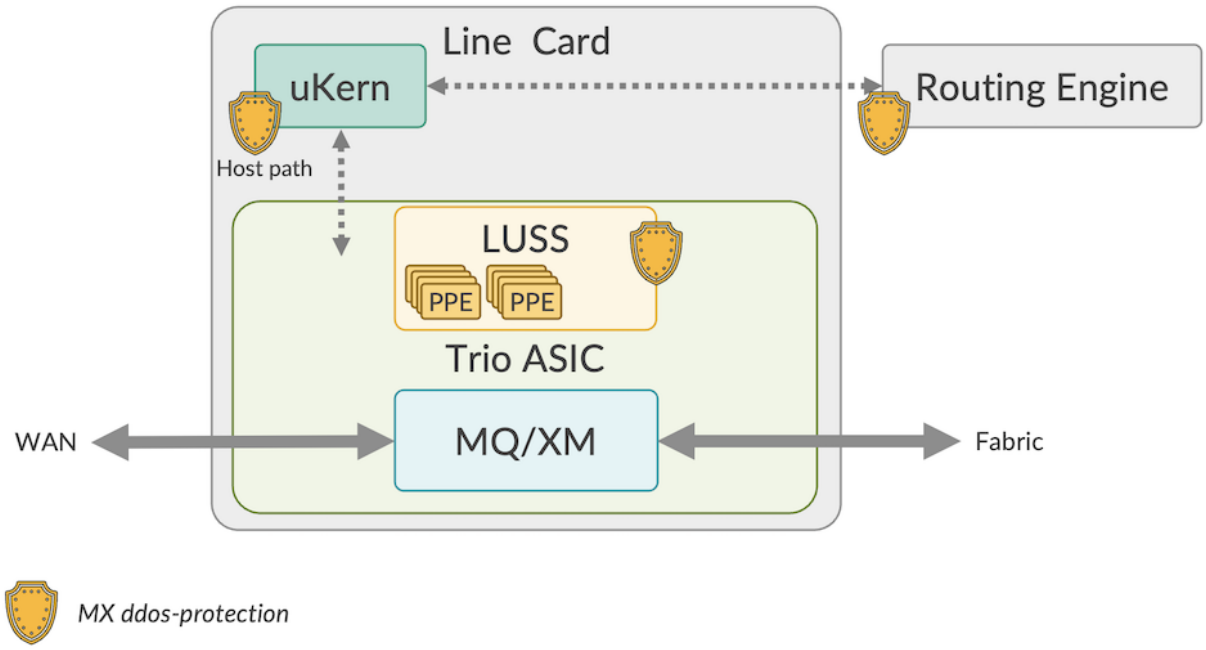


Figure 1: MX ddos-protection at 3 levels

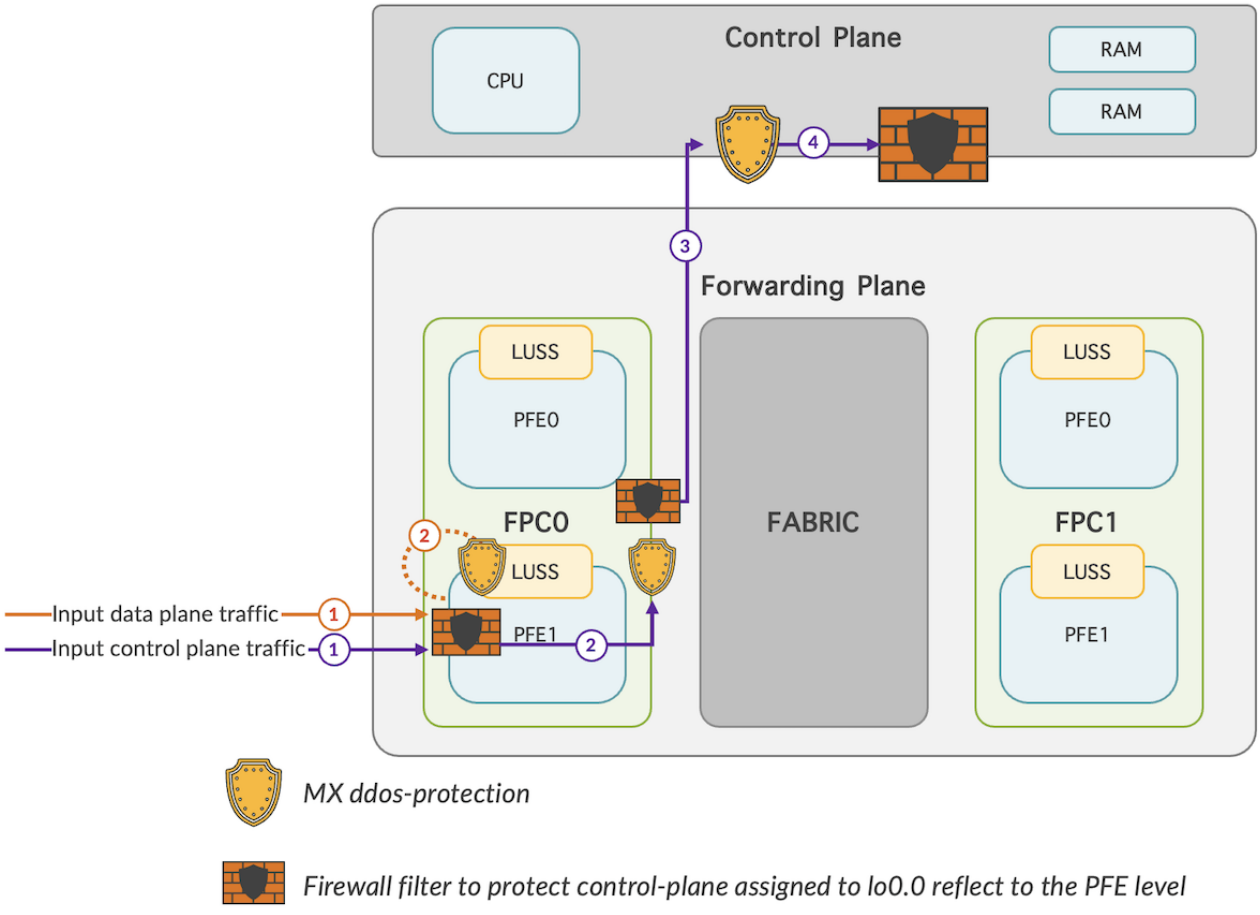


Figure 2: MX ddos-protection at 3 levels

Exception traffic will traverse through several protection mechanism before it punts to RE.

In Figure 3 below, the depicted scenario involves incoming traffic from FPC0 destined for an interface on FPC1, with the path traversing through the fabric:

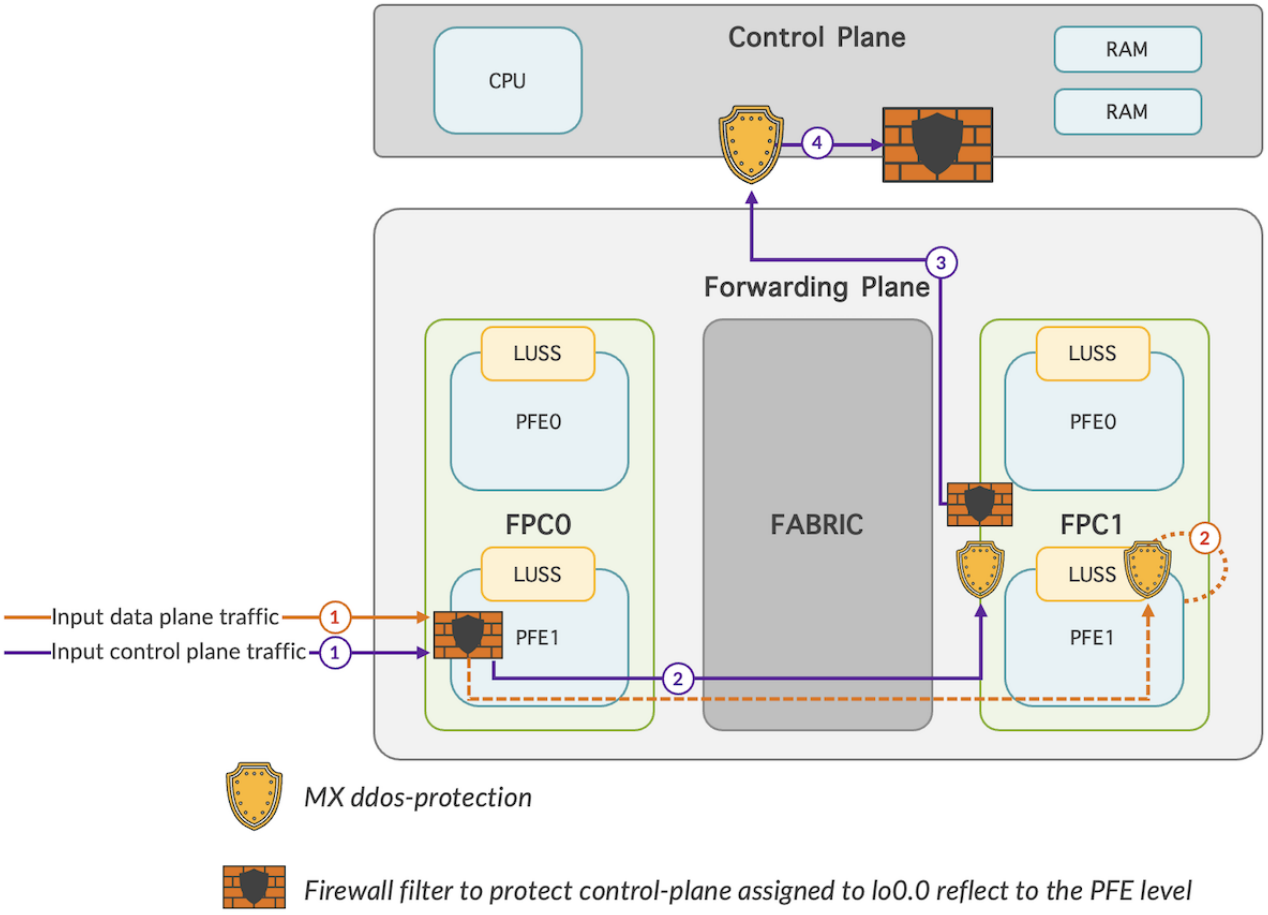


Figure 3: MX ddos-protection cross fabric

1.2 Problem Examples

1.2.1 Scenario 1

VRF attack to a protocol that is uncovered in lo0.0, associated to VRF-A

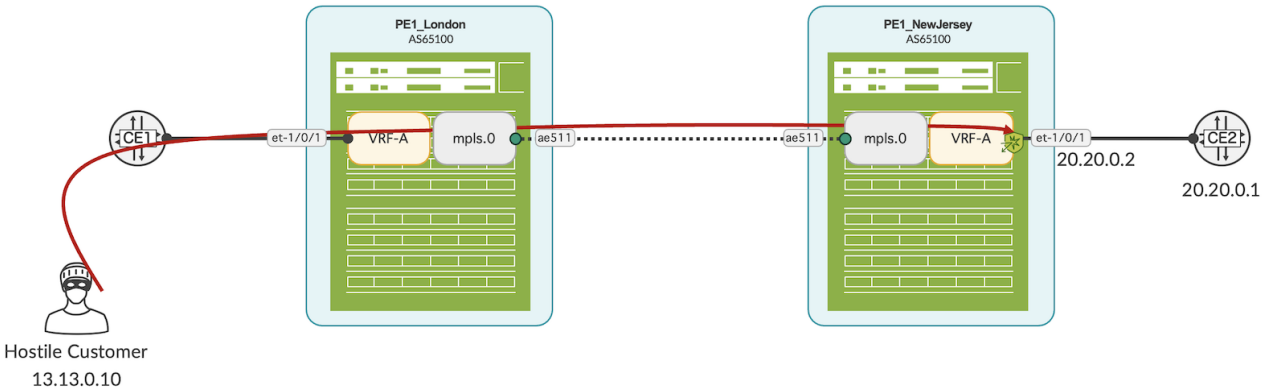


Figure 4: Scenario1

1.2.2 Scenario 2

A new loopback interface Lo0.101 has been assigned to VRF-A lacks appropriate filter protection. Within this context, a VRF attack targets an exception traffic towards either Lo0.101 or a customer facing interface.

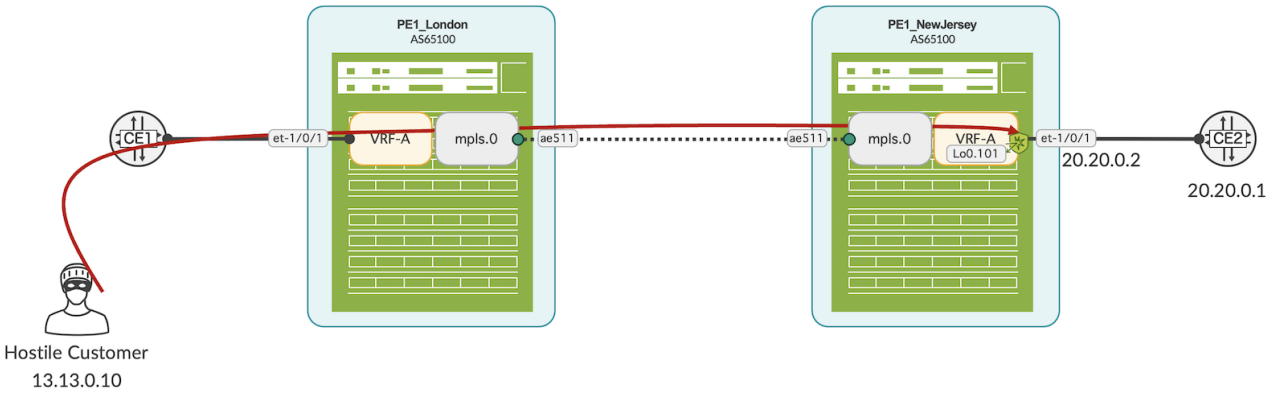


Figure 5: Scenario2

1.3 Solution Description

1.3.1 RE and VRF Filter

While this subject lies beyond the purview of this blog, its significance as a fundamental practice prompted its inclusion in this discussion. RE protection is important and required on every loopback interface which can be added separately or by using apply-groups feature.

Furthermore, other methods, including apply-path, are elaborated in the Day One book, accessible as a PDF: https://supportportal.juniper.net/s/article/Securing-the-Routing-Engine-on-M-MX-and-T-Series?language=en_US

Another method would be VRF filter:

```
set routing-instances VRF-A forwarding-options family inet filter input FW_VRF
```

With the focus of this article being MX DDoS protection, let's turn our attention back to:

1.3.2 SCFD

MX ddos-protection in SCFD (Suspicious Control Flow Detection) mode

Configure:

```
[edit]
mnayman@PE2# set system ddos-protection global flow-detection
```

Example of a BFD attack:

```
mnayman@PE2> show ddos-protection protocols violations
Packet types: 246, Currently violated: 1

Protocol Packet Bandwidth Arrival Peak Policer bandwidth
group type (pps) rate(pps) rate(pps) violation detected at
bfd aggregate 20000 1 30049 2023-07-24 23:13:26 EDT
        Detected on: FPC-3
```

```
mnayman@PE2> show ddos-protection protocols culprit-flows
Currently tracked flows: 1, Total detected flows: 2955

Protocol Packet Arriving Source Address
group type Interface MAC or IP
bfd aggregate ae1.0 10.144.0.208
  sub:00030000000006c9 2023-07-24 23:14:35 EDT pps:1 pkts:85

mnayman@PE2> show ddos-protection protocols culprit-flows detail
Currently tracked flows: 1, Total detected flows: 2957

Protocol Packet Arriving Aggr Flow Id
group type Interface level
bfd aggregate lsi.411 sub 00030000000006ca
  Source Address: 13.13.0.10
  Destination Address: 20.20.0.2
  Source Port: 3784 Destination Port: 3784
  Found at: 2023-07-24 23:15:44 EDT
  Last Violation: 2023-07-24 23:15:46 EDT
  Rate: 0 pps received packets: 165389
```

Now, what happens with the same attack coming from multiple sources:

```
mnayman@PE2> show ddos-protection protocols culprit-flows
Currently tracked flows: 1000, Total detected flows: 3969

Protocol Packet Arriving Source Address
group type Interface MAC or IP
uncls host-rt-v4 lsi.411 13.13.0.95
  sub:00030000000006d8 2023-07-24 23:22:27 EDT pps:235 pkts:5914
uncls host-rt-v4 lsi.411 13.13.1.95
  sub:00030000000006d9 2023-07-24 23:22:27 EDT pps:235 pkts:5915
uncls host-rt-v4 lsi.411 13.13.2.154
  sub:00030000000006da 2023-07-24 23:22:27 EDT pps:235 pkts:5907
uncls host-rt-v4 lsi.411 13.13.2.220
  sub:00030000000006db 2023-07-24 23:22:27 EDT pps:235 pkts:5915
uncls host-rt-v4 lsi.411 13.13.0.132
  sub:00030000000006dc 2023-07-24 23:22:27 EDT pps:235 pkts:5912
...
uncls host-rt-v4 lsi.411 13.13.2.92
  sub:0003000000000abf 2023-07-24 23:22:27 EDT pps:235 pkts:5927
```

And this is where the brilliance of the MX router shines. When facing a massive attack from thousands of sources, it initiates mitigation on a per-flow basis:

```
mnayman@PE2> show ddos-protection protocols culprit-flows
Currently tracked flows: 3036, Total detected flows: 15970

Protocol Packet Arriving Source Address
group type Interface MAC or IP
uncls host-rt-v4 lsi.411 13.13.15.63
  sub:0003000000002dc5 2023-07-24 23:25:05 EDT pps:0 pkts:185
uncls host-rt-v4 lsi.411 13.13.3.82
  sub:0003000000002dc6 2023-07-24 23:25:05 EDT pps:0 pkts:183
uncls host-rt-v4 lsi.411 13.13.5.208
  sub:0003000000002dc7 2023-07-24 23:25:05 EDT pps:0 pkts:187
uncls host-rt-v4 lsi.411 13.13.15.47
  sub:0003000000002dc8 2023-07-24 23:25:05 EDT pps:0 pkts:184
uncls host-rt-v4 lsi.411 13.13.18.20
  sub:0003000000002dc9 2023-07-24 23:25:05 EDT pps:0 pkts:185
```

The MX platform intelligently condenses the attack into a single flow, automatically assigning it one Flow ID. This resource-saving approach is enabled due to the attack's consistent signature:

```
mnayman@PE2> show ddos-protection protocols culprit-flows
Currently tracked flows: 1, Total detected flows: 17933


Protocol Packet Arriving Source Address
group type Interface MAC or IP
uncls host-rt-v4 lsi.411 -- -- --
  ifl:0003000000003e36 2023-07-24 23:25:20 EDT pps:234964 pkts:14609202

mnayman@PE2> show ddos-protection protocols culprit-flows detail
Currently tracked flows: 1, Total detected flows: 17933
```

```
12
13
14  Protocol Packet Arriving Aggr Flow Id
15  group type Interface level
16  uncls host-rt-v4 lsi.411 ifl 0003000000003e36
17    Found at: 2023-07-24 23:25:20 EDT
18    Last Violation: 2023-07-24 23:29:43 EDT
19    Rate: 234967 pps received packets: 61602109
```

Most deployments don't require any specific tweaking of the feature.

Telemetry models are also available as described in https://www.juniper.net/documentation/us/en/software/junos/open-config/interfaces-telemetry/topics/concept/junos-telemetry-interface-grpc-sensors.html

Quoting the config guide: `/junos/system/linecard/ddos/`

Distributed denial of service (DDoS) sensor. This sensor supports the Openconfig data model *junos/ui/open-config/yang/* and *junos-ddos.yang*.

You can stream information using Juniper proprietary gRPC or UDP (native) export. There are 45 packet types for DDoS. To maintain a reasonably sized data stream, data is exported for all protocols that have seen traffic using the zero-suppression model.

You can also add the following leaves to the end of the path to stream specific statistics:

- group_name
- group_id
- protocol_name
- protocol_id
- location
- received
- arrive-policer
- dropped-individual_policer
- dropped_aggregate_policer
- total_dropped
- final_passed
- arrival_rate
- max_arrival_rate
- pass_rate
- policer_state
- policer_violation_count
- policer_violation_start_time
- policer_violation_end_time
- policer_violation_duration

The following packet types are supported:

```
1  ------ --------- -------- -------- ------------------------------
2  CMICQ Channel bwidth burst Qlen Proto(s)
3  ------ ---------- -------- -------- --------- -------------------
4      0 3 500 10 200 uncls
5      4 1 4000 200 200 vchassis
6      7 3 500 200 200 vxlan
7      8 3 1500 200 200 localnh
8      9 3 1000 200 200 vcipc-udp
9     10 3 2000 200 200 sample-source
10    11 3 2000 200 200 sample-dest
11    12 3 50 10 200 l3mtu-fail,ttl,ip-opt.
```

```
    14 3 100 10 200 garp-reply
    15 3 500 10 200 fw-host
    16 3 500 200 200 ndpv6
    17 3 1000 200 200 dhcpv4v6
    19 3 1500 200 200 ipmc-reserved
    20 3 300 200 200 resolve
    21 3 100 10 200 l3dest-miss
    22 3 100 10 200 redirect
    23 3 300 200 200 l3nhop
    24 3 100 10 200 l3mc-sgvhit-icl
    25 3 50 10 200 martian-address
    26 3 1000 200 200 l2pt
    27 3 50 10 200 urpf-fail
    28 3 1000 300 300 ipmcast-miss
    29 2 300 10 200 nonucast-switch
    30 2 3000 200 200 rsvp,ldp,bgp
    31 2 3000 200 200 unknown-l2mc,rip,ospf
    32 2 1000 200 200 fip-snooping
    33 2 1000 200 200 igmp
    34 2 500 200 200 arp
    35 2 1500 200 200 pim-data
    36 2 1500 200 200 ospf-hello
    37 2 1500 200 200 pim-ctrl
    38 2 2000 200 200 isis
    39 1 250 200 200 lacp
    40 1 1200 200 200 bfd
    41 1 100 10 200 ntp
    42 1 500 200 200 vchassis
    43 1 1000 200 200 stp,pvstp,lldp
```

1.4 Useful links

- Control Plane DDoS Protection Flow Detection Overview: https://www.juniper.net/documentation/us/en/software/junos/security-services/topics/concept/subscriber-management-scfd-overview.html
- MX Series, 2nd Edition Chapter 4: https://www.oreilly.com/library/view/juniper-mx-series/9781491932711/ch04.html
- Telemetry config guide: https://www.juniper.net/documentation/us/en/software/junos/open-config/interfaces-telemetry/topics/concept/junos-telemetry-interface-grpc-sensors.html

1.5 Glossary

- AS: Autonomous System
- BFD: Bidirectional Forwarding Detection
- DDOS: Distributed Denial-of-Service
- FPGA: Field Programmable Gate Array
- IP: Internet Protocol
- Junos: Operating System used in Juniper Networks routing, switching and security devices.
- MPLS: Multiprotocol Label Switching
- MX: Multi-Service router with programmable silicon
- PE: Provider Edge router
- PFE: Packet Forwarding Engine
- PIC: Physical Interface Cards
- RE: Routing Engine
- SCFD: Suspicious Control Flow Detection
- Trio: Juniper silicon. multi-threaded programmable packet processing engine and a hierarchy of high-capacity memory systems
- VPN: Virtual Private Network



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